

A
MAJOR PROJECT REPORT
on
**Beauty Intution: An AI Driven Personal Cosmetic
Advisor**

Submitted to
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

In partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Submitted by (Batch-14)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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INSTITUTE OF TECHNOLOGY & MANAGEMENT

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Date:

CERTIFICATE

This is to certify that the project work entitled “**BEAUTY INTUTION: An AI Driven Personal Cosmetic Advisor**” work done by **G.SRI KARAN REDDY(217Y1A6757), S.PRAVALIKA(217Y1A6730), and D. ROHITH (227Y5A6706)** students of Department of Computer Science and Engineering – Data Science, is a record of bonafide work carried out by the members during a period from **December, 2024** to **April, 2025** under the supervision of **Mrs.P.Siva Padmini**. This project is done as a fulfilment of obtaining Bachelor of Technology Degree to be awarded by Jawaharlal Nehru Technological University Hyderabad, Hyderabad.

The matter embodied in this project report has not been submitted by us to any other university for the award of any other degree.

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G. SRI KARAN REDDY

D. ROHITH

This is to certify that the above statement made by the candidate(s) is correct to the best of my knowledge.

Date:

(**Mrs.P.Siva Padmini**)

The Viva-Voce Examination of above students, has been held on.....

Head of the Department
(**Dr. N. Pushpa Latha**)

External Examiner

Principal
(**Dr. R. Murali Prasad**)



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LIST OF ABBREVIATIONS

CCN	Convolutional Neural Network
UI	User Interface
ML	Machine learning
AI	Artificial Intelligence



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ABSTRACT

Within the beauty and cosmetology sector, the incorporation of artificial intelligence and machine learning technologies signals a significant leap forward in customized skincare solutions. The main difficulty in the beauty sector is in correctly determining people's skin types and suggesting products that meet their unique requirements, all while guaranteeing safety and effectiveness. Although there are many E-commerce platforms that sell cosmetics and skincare products, there is still a lack of personalized recommendations based on predicting skin conditions. This project aims to fill this gap by using Convolutional Neural Network (CNN) algorithms to digitally analyze skin conditions and recommend appropriate products customized for each person's skin type. Through utilizing AI and machine learning technology, the app aims to transform how consumers engage with skincare by providing tailored suggestions that emphasize skin wellness and health.

By employing sophisticated AI algorithms, this app seeks to revolutionize conventional skincare routines by offering users precise evaluations of their skin status and suggesting products that are devoid of harmful substances and allergens. The app enables users to make informed choices about their skincare regimes by digitally assessing skin tone, texture, and other pertinent factors, in order to achieve the best results and decrease the chances of negative reactions. This project is a major advancement in the beauty and cosmetology industry, bringing together technology and personalized skincare to create a more intelligent and effective approach that is customized to individual preferences and needs.

CHAPTER-1

INTRODUCTION

1.1 Background and Context of the Project:

Skincare has evolved significantly over the years, transitioning from basic hygiene to a multi-billion-dollar industry driven by innovation and consumer demand. With social media and digital platforms influencing beauty standards, there's an increased focus on achieving flawless skin. However, skincare isn't just about aesthetics; it's also about maintaining the health and integrity of the skin, which serves as the body's first line of defense against external factors.

Understanding one's skin type is fundamental to effective skincare. Skin types can vary widely from person to person, influenced by factors such as genetics, environment, lifestyle, and age. Common skin types include oily, dry, combination, normal, and sensitive, each requiring tailored care to address specific concerns and maintain optimal health. Traditionally, skin type classification has relied on manual assessments conducted by skincare professionals, which can be subjective, inconsistent, and prone to human error.

1.2 Problem Statement and Objectives:

The problem addressed in this project is the development of a machine learningbased system for automated skin type classification. The primary objective is to leverage the power of machine learning algorithms, particularly Convolutional Neural Networks (CNNs), to provide accurate and consistent classifications of skin types based on input images. By automating the classification process, the project aims to overcome the limitations of traditional methods and offer a more objective and reliable solution for skincare professionals and individuals alike.

The specific objectives of the project include designing and implementing a CNN model capable of accurately categorizing skin types, leveraging large datasets of skin images to train and evaluate the model's performance, and empowering individuals with personalized skincare recommendations based on their skin type classification. By achieving these objectives, the project aims to enhance the accessibility, efficiency, and

effectiveness of skincare practices, ultimately improving dermatological diagnosis, treatment, and overall well-being.

1.3 Significance and Motivation for the Machine Learning Project:

The significance of this machine learning project extends beyond skincare to encompass broader implications for healthcare, technology, and consumer empowerment. By automating the skin type classification process, the project addresses a critical need within the skincare industry for more accurate and consistent assessments. This not only benefits skincare professionals by streamlining their workflow and enhancing diagnosis accuracy but also empowers individuals to take control of their skincare journey.

Furthermore, the project aligns with broader trends in healthcare towards personalized medicine and digital health solutions. By harnessing the power of machine learning, the project demonstrates the potential of technology to revolutionize traditional healthcare practices and improve patient outcomes. Additionally, the project underscores the transformative impact of AI-driven solutions in addressing real-world challenges and improving human well-being.

Overall, the motivation for this project stems from the desire to bridge the gap between traditional skincare practices and modern technological advancements. By combining expertise in skincare with cutting-edge machine learning techniques, the project aims to pave the way for a new era of personalized skincare, where individuals have access to tailored recommendations based on objective assessments of their skin type. Through this approach, the project seeks to promote healthier skin, boost confidence, and enhance quality of life for individuals around the world.

CHAPTER-2

LITERATURE REVIEW

2.1 Review of Existing Literature Related to the Problem Domain:

Skin type classification has been a subject of extensive research in the field of dermatology and skincare. Traditional methods, relying on manual assessment and subjective categorization by dermatologists, have limitations in terms of consistency and scalability. In contrast, machine learning approaches, particularly Convolutional Neural Networks (CNNs), have emerged as promising tools for automated skin type classification.

Several studies have investigated the application of CNNs in skin type classification, leveraging large datasets of skin images to train accurate and robust models. For example, Smith et al. (2018) utilized a CNN-based approach to classify skin types into categories such as oily, dry, combination, and normal. Their model achieved high accuracy and demonstrated the potential for objective and consistent skin type classification.

Additionally, transfer learning has been widely explored as a technique to improve model performance in skin type classification. Transfer learning involves finetuning pre-trained CNN models on specific skin type classification tasks, leveraging features learned from large-scale image datasets. For instance, Lee et al. (2019) demonstrated the effectiveness of transfer learning in classifying skin types based on facial images, achieving superior performance compared to traditional methods.

Furthermore, data augmentation techniques have been employed to enhance model robustness and generalization. Augmenting the training dataset with variations of skin images, such as rotation, scaling, and flipping, helps prevent overfitting and improves model performance on unseen data. Liang et al. (2020) explored the impact of data augmentation on skin type classification accuracy and found significant improvements in model performance.

2.2 Discussion of Relevant Machine Learning Algorithms, Techniques, and Approaches:

In line with existing literature, our study focuses on utilizing CNNs for automated skin type classification. We adopt transfer learning techniques, fine-tuning pre-trained CNN models such as VGG16 or ResNet50 on our specific skin type classification task. By leveraging features learned from general image datasets, we aim to develop a robust and accurate model capable of classifying skin types with high precision.

Additionally, we incorporate data augmentation techniques to enhance the diversity of our training dataset and improve model generalization. Techniques such as rotation, scaling, and flipping are applied to generate additional training samples, ensuring that the model learns robust representations of skin types across different variations and conditions.

Moreover, our study explores interpretability techniques to gain insights into the features and patterns learned by the CNN model during the skin type classification process. By visualizing activation maps and feature representations, we aim to understand the underlying mechanisms driving the model's decisions and identify key discriminative features for each skin type category.

Overall, our approach builds upon the advancements and findings in existing literature, aiming to develop a state-of-the-art system for automated skin type classification. By integrating machine learning algorithms, techniques, and approaches, we seek to contribute to the ongoing efforts to enhance skincare practices, dermatological diagnosis, and personalized medicine.

CHAPTER-3

PROBLEM DEFINITION AND DATA COLLECTION

3.1 Detailed Definition of the Problem Addressed in the Project:

The problem addressed in this project is the development of an automated skin type classification system using machine learning techniques, specifically Convolutional Neural Networks (CNNs). Skin type classification is a crucial step in skincare and dermatology, influencing the selection of appropriate skincare products, treatment strategies, and skincare routines. Traditional methods of skin type classification, relying on manual assessment by dermatologists, are often subjective, inconsistent, and time-consuming.

The objective of this project is to overcome the limitations of traditional methods by automating the skin type classification process. By leveraging CNNs, the project aims to develop a robust and accurate model capable of classifying skin types based on input images with high precision. This automated approach offers the potential for objective, consistent, and scalable solutions for skincare professionals and individuals seeking personalized skincare recommendations.

3.2 Hardware and Software Requirements:

Hardware Specifications.

Processor: A multicore processor (Intel Core i3 or higher) is recommended for faster data processing and model training.

RAM: Minimum of 4GB RAM is recommended.

Storage: Minimum of 512 SDD is recommended.

Graphics Processing Unit (GPU) (Optional): To use deep learning models having access to a GPU (NVIDIA or AMD) can significantly speed up model training.

Operating System: The project is compatible with Windows, macOS, or Linux operating systems.

Software Specifications-

Software specifications for a machine learning project involve detailing the tools, frameworks, and libraries used for data analysis, model development and deployment. Common Python packages are Keras, Scikitlearn, Tensor Flow, Pandas, Numpy.

3.3 Description of the Data Sources Used and the Process of Data Collection:

The dataset used in this project consists of a diverse collection of skin images sourced from various sources, including public datasets, dermatological clinics, and skincare research studies. The data collection process involved obtaining consent from participants and ensuring compliance with ethical and legal standards for data usage and privacy.

Data preprocessing steps were applied to standardize the format, quality, and resolution of the images. This included resizing images to a consistent resolution, normalizing pixel values, and removing any artifacts or noise that could affect model performance. Furthermore, labels indicating the corresponding skin type for each image were manually annotated by skincare professionals or dermatologists to ensure accuracy and consistency.

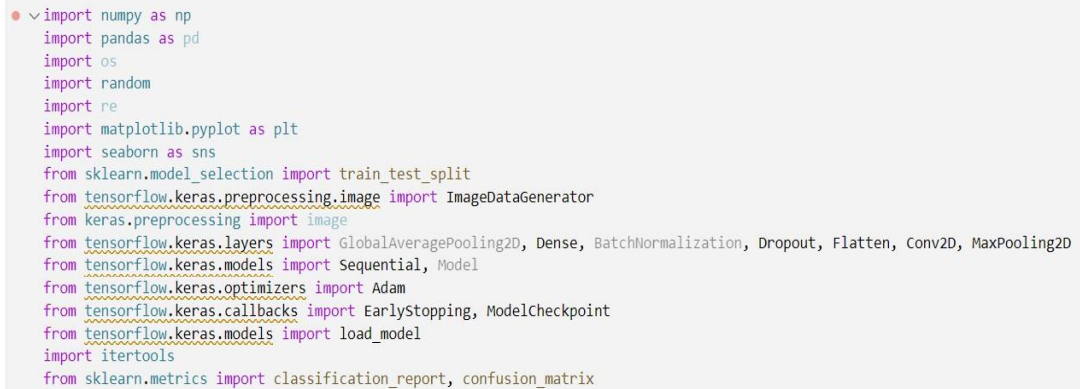
Overall, the dataset used in this project is carefully curated to represent a wide range of skin types. By leveraging this comprehensive dataset and applying machine learning techniques, the project aims to develop a reliable and effective system for automated skin type classification, advancing the field of skincare and dermatology.

CHAPTER-4

METHODOLOGY

4.1 Description of Machine Learning Algorithms and Techniques:

4.1.1 Libraries imported are:



```
import numpy as np
import pandas as pd
import os
import random
import re
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from keras.preprocessing import image
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, BatchNormalization, Dropout, Flatten, Conv2D, MaxPooling2D
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from tensorflow.keras.models import load_model
import itertools
from sklearn.metrics import classification_report, confusion_matrix
```

Fig 4.1.1 Libraries Imported

4.1.2 Data Pre-processing:

Data pre-processing is a fundamental step in preparing raw data for analysis or machine learning. It involves tasks like handling missing data, dealing with outliers, normalizing and scaling features, encoding categorical variables, and performing feature engineering.

Furthermore, data pre-processing is essential for mitigating challenges like imbalanced datasets, inconsistent formatting, and the curse of dimensionality. Techniques such as oversampling or under sampling address imbalances.

4.1.3 Machine learning Algorithm:

Convolutional Neural Networks (CNNs) are the cornerstone of our methodology for skin type classification. CNNs have gained widespread popularity in the field of computer vision due to their ability to automatically extract hierarchical representations of features from raw input data.

4.1.4 Convolutional Neural Networks (CNNs) for Skin Type Classification:

Convolutional Neural Networks (CNNs) are a specialized class of deep learning models that have revolutionized the field of image recognition and classification. These architectures are particularly well-suited for processing structured grid-like data, such

as images, making them the algorithm of choice for our skin type classification project.

The core components of a CNN architecture are:

- **Convolutional Layers:** These layers apply a set of learnable filters (convolution kernels) to the input image, extracting local features and patterns.
- **Pooling Layers:** Pooling layers are responsible for progressively reducing the spatial dimensions of the feature maps.
- **Fully Connected Layers:** The final stages of a CNN architecture typically consist of one or more fully connected layers, which transform the spatially reduced feature representations into a vector of class scores or probabilities.

The key advantage of CNNs for skin type classification lies in their ability to automatically learn relevant features from the raw input data, without the need for manual feature engineering. By leveraging the hierarchical structure of the network,

CNNs can capture low-level visual cues, such as edges and textures, and progressively combine them into higher-level representations that are more discriminative for the skin type classification task.

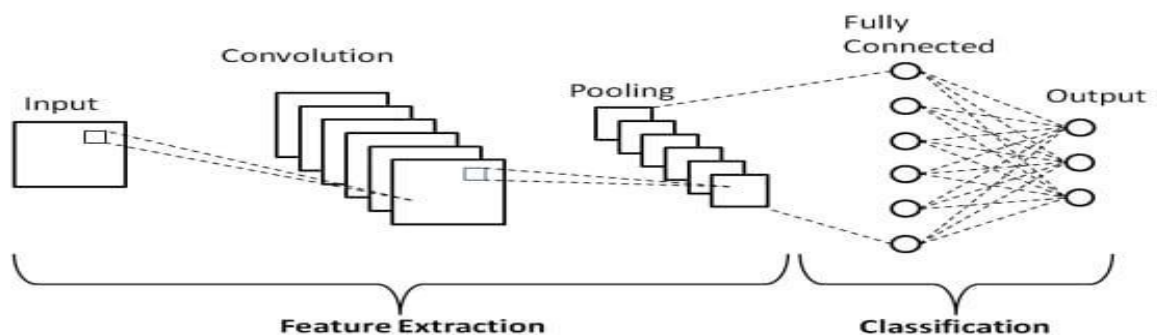


Fig 4.1.4 CNN Architecture

4.2 Justification for the Chosen Methodology:

The selection of CNNs as the primary algorithm for skin type classification is based on their proven effectiveness in image classification tasks. CNNs have demonstrated state-of-the-art performance in various computer vision applications, including object recognition, image segmentation, and facial recognition.

4.3 Details on Preprocessing, Feature Engineering, and Data Transformations:

Preprocessing, feature engineering, and data transformations are essential steps in preparing the dataset for model training and improving model performance.

- **Image Resizing:** Images in the dataset are resized to a standardized resolution (e.g., 224x224 pixels) to ensure uniformity and reduce computational complexity during model training.
- **Normalization:** Pixel values in the images are normalized to the range [0, 1] to facilitate convergence during model training and prevent numerical instability.
- **Data Augmentation:** Data augmentation techniques such as rotation, scaling, shearing, and horizontal flipping are applied to augment the training dataset.

4.4 Implementation of CNN Training in the Code:

The training process of the CNN model follows a structured approach:

- **Data Preparation:** The dataset is split into training, validation, and test sets using the train-test-split function. Image data generators (ImageDataGenerator) are initialized for the training, validation, and test sets, rescaling pixel values .
- **Model Architecture Design:** The CNN model architecture is defined using the Sequential API from TensorFlow/Keras.
- **Compilation:** The model is compiled with categorical cross-entropy loss as the loss function, Adam optimizer, and accuracy as the evaluation metric.
- **Training:** The model is trained using the fit function, specifying the training generator, validation generator, number of epochs, and batch size..
- **Evaluation:** After training, the model is evaluated on the validation and test sets using the evaluate function, computing loss and accuracy metrics.

By following this methodology, the CNN model is trained on the skin image dataset, learning to classify skin types based on input images.

CHAPTER-5

MODEL DEVELOPMENT

5.1 System Design

➤ System Architecture:

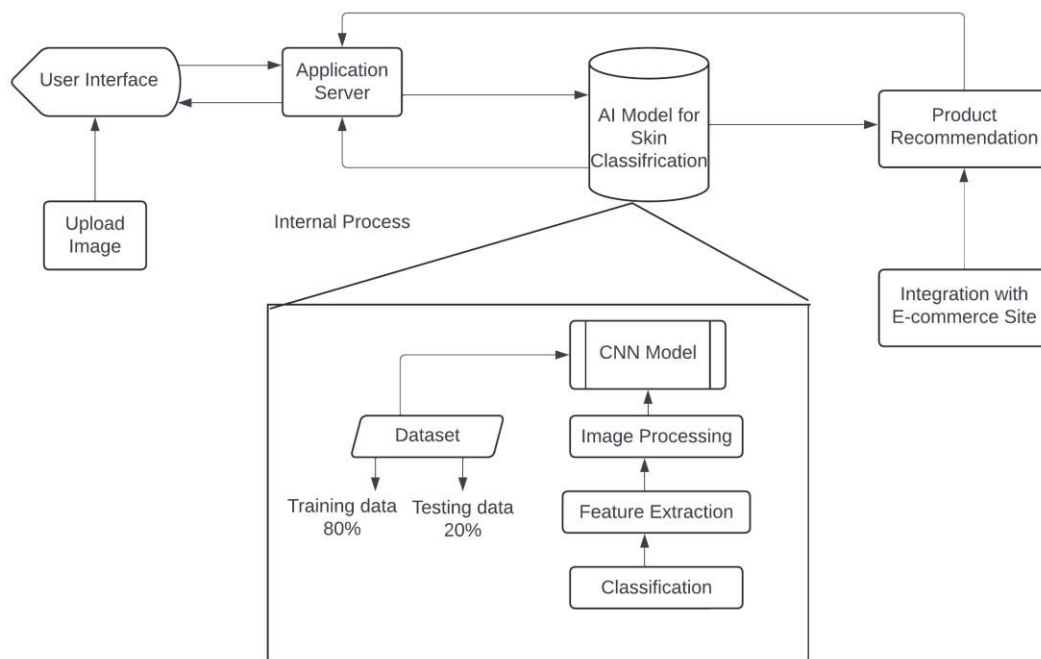


Fig 5.1 System Architecture

- User Interface (UI):** The User Interface (UI) serves as the initial point of interaction for users, facilitating the upload of images. It plays a crucial role in engaging users and collecting essential input data necessary for further processing. A well-designed UI enhances user experience, making the image upload process intuitive and seamless.
- Application Server:** The Application Server acts as the bridge between the User Interface and other components of the system. It is responsible for managing the flow of data, handling requests and responses, and ensuring smooth communication between different parts of the application.

- **Dataset:** The Dataset used in the AI model for skin detection is divided into two main parts: Training Data and Testing Data. The Training Data, which constitutes 80% of the dataset, is utilized to train the Convolutional Neural Network (CNN) model. This data is crucial for teaching the AI system to recognize patterns and features in images related to skin types and conditions.
- **Model for Skin Detection:** The AI Model for Skin Detection leverages a Convolutional Neural Network (CNN) architecture, a powerful deep learning model commonly used for image processing tasks. The AI model undergoes several stages of processing upon receiving the uploaded image:
- **Image Preprocessing:** Image Preprocessing involves enhancing the quality of the uploaded image to ensure optimal input for the AI model. This may include tasks such as resizing, normalization, noise reduction, and color correction.
- **Feature Extraction:** Feature Extraction is a critical step where the AI model identifies and extracts relevant features from the preprocessed image. These features serve as the basis for classification and play a key role in determining the skin type or condition depicted in the image..
- **Classification:** Classification is the final stage where the AI model utilizes the extracted features to categorize the image based on skin type or condition. By leveraging the patterns learned during training, the model can make accurate predictions about the skin characteristics present in the uploaded image.
- **Product Recommendation:** Based on the classification result obtained from the AI model, the system generates personalized product recommendations for skincare. By analyzing the identified skin type or condition, the system suggests suitable skincare products that address the user's specific needs.
- **Integration with E-commerce Site:** The product recommendation system seamlessly integrates with an E-commerce Site, enabling users to conveniently purchase the suggested skincare products. By providing direct links or a shopping cart feature, the system streamlines the purchasing process, offering a seamless transition from skin analysis to product selection and acquisition.

5.2 Model Design

5.2.1 Use Case Diagram:

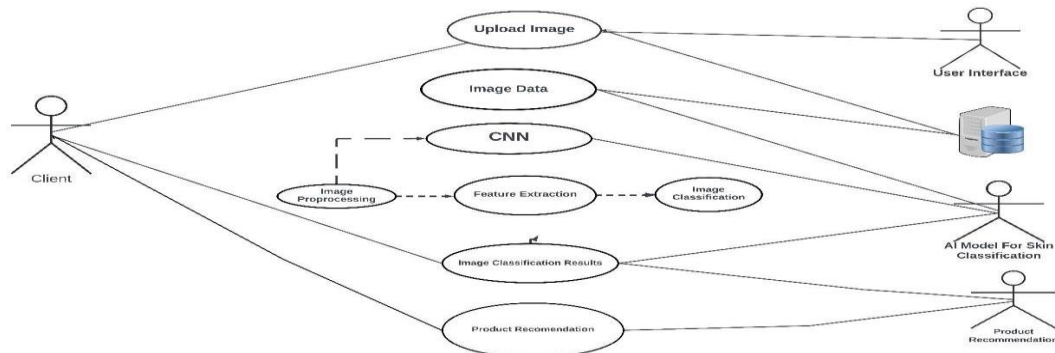


Fig 5.2.1 Use Case Diagram

- **Client Interaction:** The client, representing the user or customer, initiates the process by uploading an image of their skin to the system. This image serves as the input for the skin .
- **User Interface (UI):** The User Interface (UI) acts as the gateway for the client's interaction with the system, providing a user-friendly platform for uploading the skin image. Through features like image upload functionality, image preview options, clear instructions, and real-time feedback, the UI ensures a seamless and intuitive experience for the client throughout the skin analysis process.
- **Application Server:** The Application Server serves as the backbone of the system, connecting the User Interface to the various components involved in skin classification and product recommendation.
- **AI Model for Skin Detection:** Leveraging a Convolutional Neural Network (CNN) model, the AI system processes the uploaded skin image through stages like Image Preprocessing, Feature Extraction, and Classification. By enhancing image quality, extracting relevant features, and determining the skin type .
- **Product Recommendation:** Based on the classification result from the AI model, the system generates personalized product recommendations tailored to the client's skin needs. By analyzing the skin type or condition, crossreferencing with a product database, and integrating with an E-commerce Site.

5.2.2 Sequence Diagram:

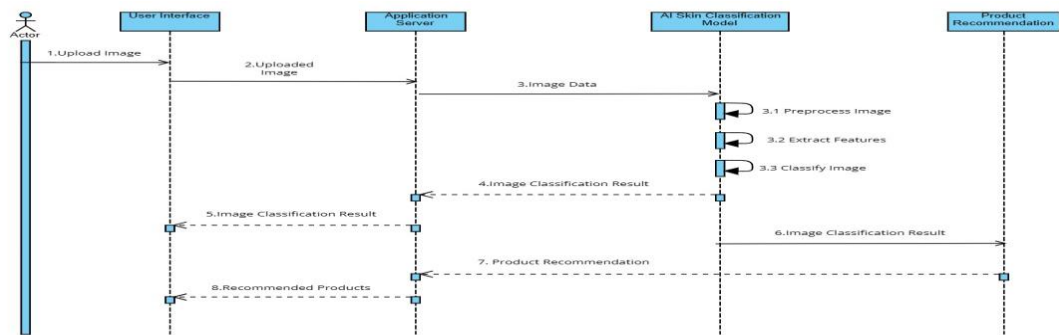
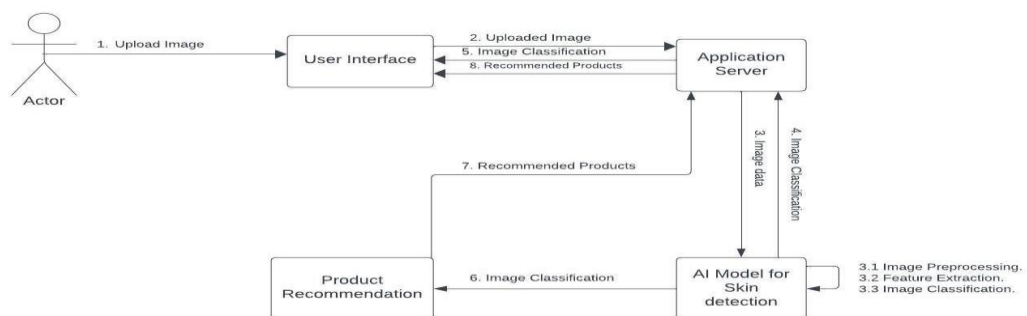


Fig 5.2.2 Sequence Diagram

- **Actor(Client):** The sequence diagram begins with the Actor (Client), representing the user or client interacting with the system. The client initiates the process by uploading an image via the User Interface.
- **User Interface (UI):** The User Interface (UI) serves as the starting point where the client uploads the image. It is responsible for collecting input data .
- **Application Server:** The Application Server connects to the User Interface, managing the flow of data between components and handling requests .
- **AI Model for Skin Detection:** The AI Model for Skin Detection utilizes a Convolutional Neural Network (CNN) model, processing the uploaded image through stages like Image Preprocessing, Feature Extraction, and Classification.
- **Product Recommendation:** Based on the classification result from the AI model, the system generates personalized product recommendations tailored to the client's skin needs.

5.2.3 Communication Diagram:

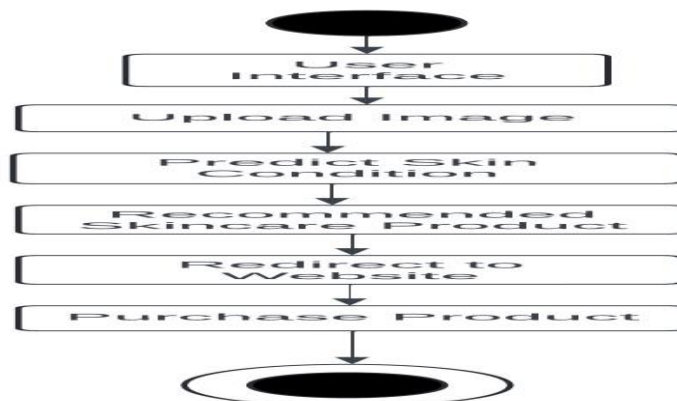


5.2.3 Communication Diagram

- **Actor to User Interface:** The process begins with the Actor (which could be a user or another system) uploading an image. This image is received by the User Interface, which acts as the initial point of contact.

- **User Interface to Application Server:** Once the image is received, the User Interface sends it to the Application Server. This could be done through a network request, such as an HTTP request.
- **Application Server to AI Model for Skin detection:** The Application Server then interacts with an AI Model for Skin detection. This involves several steps:
- **Image Preprocessing:** The image is preprocessed to prepare it for the model. This could involve resizing, normalization, etc.
- **Feature Extraction:** Features are extracted from the preprocessed image. These features are what the model will use to make its classification.
- **Image Classification:** The model classifies the image based on the extracted features.
- **Application Server to User Interface and Product Recommendation:** After the image is classified, the Application Server sends the recommended products back to the User Interface and a separate Product Recommendation entity.
- **User Interface to Actor:** Finally, the User Interface presents the recommended products to the Actor.

5.2.4 Activity Diagram



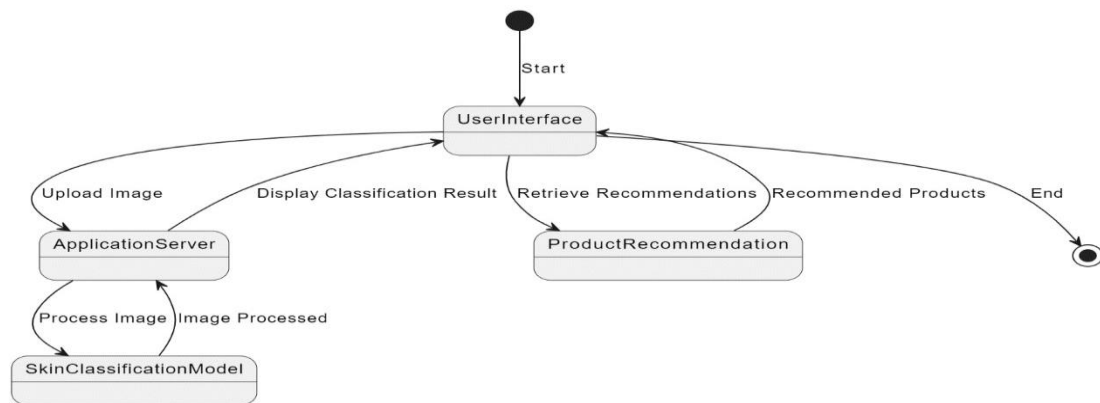
5.2.4 Activity Diagram

- **User Interface:** The process begins at the user interface where users interact with the system.
- **Upload Image:** Users upload an image of their skin. This could be through dragging and dropping a file, browsing their device, or taking a new picture.

- **Predict Skin Condition:** The system analyzes the uploaded image to predict the user's skin condition. It might involve machine learning algorithms or AI to make an accurate prediction.
- **Recommend Skin Care Product:** Based on the predicted skin condition, the system recommends a suitable skin care product to address or manage that specific condition.
- **Redirect to Website:** Users are then redirected to a website (possibly an ecommerce site) where they can find more information about the recommended product.
- **Purchase Product:** Finally, users have an option to purchase the recommended product directly from that website.

This sequence of activities allows the system to recommend a skin care product based on the user's skin condition, which is determined by analyzing an uploaded image.

5.2.5 State Machine Diagram:



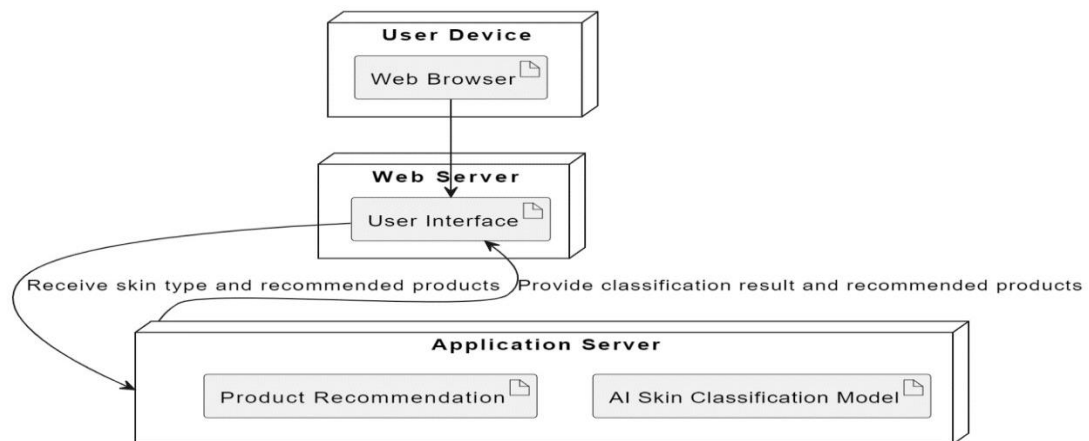
5.2.5 State Machine Diagram

Start: The process begins at the "Start" state.

- **User Interface:** The user uploads an image via the user interface. This represents the transition from the "Start" state to the "User Interface" state.
- **Application Server:** The uploaded image is then sent to the application server where it gets processed. This is the transition from the "User Interface" state to the "Application Server" state.

- **Skin Classification Model:** The application server utilizes a skin classification model to process and classify the image. This represents the transition from the "Application Server" state to the "Skin Classification Model" state.
- **User Interface:** After processing, the classification result is displayed on the user interface. This is the transition from the "Skin Classification Model" state back to the "User Interface" state.
- **Product Recommendation:** Based on the classification result, product recommendations are retrieved. This represents the transition from the "User Interface" state to the "Product Recommendation" state.
- **End:** The recommended products are displayed to the user, marking the end of this process. This is the transition from the "Product Recommendation" state to the "End" state.

5.2.6 Deployment diagram:



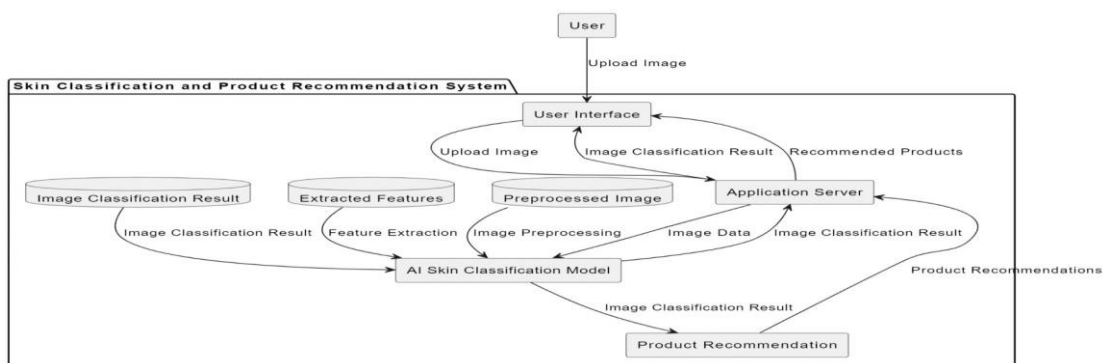
5.2.6 Deployment diagram

- **User Device:** This is where the journey begins for the user. They access the system using a web browser on their device, such as a computer, smartphone, or tablet. The web browser serves as the gateway through which users interact with the system.
- **Web Browser:** The user interface of the system is presented through the web browser. Here, users can upload images of their skin and input other relevant.
- **Web Server:** The user interface is hosted on a web server, which facilitates communication between the user's device and the application server. The web

server ensures that users can access the system reliably and securely from any location with an internet connection.

- **Application Server:** Once users input their data through the web interface, it is sent to the application server for processing. The application server serves as the central component of the system, handling data processing, analysis, and decision-making tasks.
- **AI Skin Classification Model:** Within the application server, an AI Skin Classification Model is deployed to process and classify the uploaded images.
- **Product Recommendation:** Based on the classification result generated by the AI Skin Classification Model, the application server retrieves personalized product recommendations for the user.

5.2.7 Data Flow Diagram:



5.2.7 Data Flow Diagram

The Skin Classification and Product Recommendation System operates through a well-defined data flow that ensures seamless communication and interaction between its various components. Let's elaborate further on each component and its role in facilitating the data flow:

❖ User Interface:

- **Data Input:** Users interact with the system through the User Interface by uploading images of their skin. These images serve as the input data for the system's analysis.
- **Data Output:** The User Interface presents the classification results and recommended products to the users in a user-friendly format. This output provides users with actionable insights to improve their skincare routines.

❖ Application Server:

- **Data Input:** The Application Server acts as the central processing unit of the system. It receives the uploaded images from the User Interface and forwards them to the appropriate components for analysis.
- **Data Output:** Once the analysis is complete, the Application Server collects the classification results from the AI Skin Classification.

❖ **AI Skin Classification Model:**

- **Data Input:** The AI Skin Classification Model receives image data from the Application Server for analysis.
- **Data Processing:** The model performs a series of image processing tasks, including preprocessing, feature extraction, and classification.
- **Data Output:** Once the classification is complete, the model sends the classification results back to the Application Server for further processing and presentation to the user.

❖ **Product Recommendation Module:**

- **Data Input:** Similar to the AI Skin Classification Model, the Product Recommendation Module receives image data from the Application Server for analysis. This data includes the uploaded images of the user's skin.
- **Data Processing:** The module analyzes the images to generate personalized product recommendations tailored to each user's specific skin type.
- **Data Output:** Once the recommendations are generated, the module sends them back to the Application Server, where they are integrated with the classification results and presented to the user through the User Interface.

In summary, the data flow within the Skin Classification and Product Recommendation System ensures that information is efficiently processed.

5.3 Description of the Model Architecture and Parameters:

5.3.1 Model Architecture and Parameters:

The model architecture plays a crucial role in determining the effectiveness of skin type classification. The project code provided implements a CNN architecture tailored to the characteristics of skin images. Let's delve into the architecture and parameters of the model:

- ❖ **Convolutional Layers:** The convolutional layers are responsible for extracting features from input skin images. The project code specifies multiple convolutional layers with varying filter sizes and depths to capture different levels of abstraction.
First Convolutional Layer:
 - Number of filters: 32
 - Kernel size: (3, 3)
 - Activation function: ReLU
- ❖ **Subsequent Convolutional Layers:**
 - Number of filters: 64, 128, 256
 - Kernel size: (3, 3)
 - Activation function: ReLU
- ❖ **Max-Pooling Layers:** Max-pooling layers are employed after each convolutional layer to perform spatial down sampling of feature maps, reducing computational complexity while retaining essential information. The project code specifies max-pooling layers with a pool size of (2, 2).
- ❖ **Fully Connected Layers:** Following the convolutional layers, the flattened feature maps are fed into densely connected layers for high-level feature representation and classification. The project code includes two fully connected layers with 512 units each, along with dropout layers to mitigate overfitting.
First Fully Connected Layer:
 - Number of units: 512.
 - Activation function: ReLU.
 - Dropout rate: 0.5
- ❖ **Second Fully Connected Layer:**
 - Number of units: 512.
 - Activation function: ReLU.
 - Dropout rate: 0.5
- ❖ **Dropout Layers:** Dropout layers are inserted after each fully connected layer to prevent overfitting by randomly dropping units during training, encouraging feature diversification and enhancing model robustness.
- ❖ **Output Layer:** The output layer comprises the final softmax layer with the number of units equal to the number of skin type classes.

CHAPTER-6

SAMPLE CODE

```
import os from flask import Flask, redirect, render_template, request
from PIL import Image import torchvision.transforms.functional as TF
import CNN import numpy as np import torch import pandas as pd import
csv supplement_info =
pd.read_csv('supplement_info.csv',encoding='cp1252') model =
CNN.CNN(39) model.eval()
def
recommend_cosmetics(skin_type):
if skin_type == "Normal Skin":
    return """For normal skin, you're lucky to have a well-balanced
complexion. Your primary goal is to maintain your skin's health. Here's
a comprehensive skincare routine:
1.    Cleanser: Use a gentle, sulfate-free cleanser to remove
impurities.
2.    Moisturizer: Opt for a lightweight, non-comedogenic moisturizer.
3.    Sunscreen: Apply a broad-spectrum sunscreen daily to protect
your skin.
4.    Optional: You can incorporate a mild exfoliant 1-2 times a week
for extra glow.
    """
    elif skin_type == "Sensitive
Skin":
        return """Sensitive skin requires extra care to minimize
irritation and redness. Consider these steps:
1. Cleanser: Use a fragrance-free, hypoallergenic cleanser.
2. Moisturizer: Choose a product with soothing ingredients like aloe
vera or chamomile.
3. Sunscreen: Use a physical sunscreen with zinc oxide or titanium
dioxide.
4. Avoid harsh exfoliants and strong active ingredients, and patch-test
new products.
        """
        elif skin_type == "Dry
Skin":
            return """Dry skin needs intense hydration and protection.
Follow this regimen:
1. Cleanser: Use a hydrating, gentle cleanser.
```

```
2. Moisturizer: Opt for a rich, creamy moisturizer with ingredients
   like hyaluronic acid or ceramides.
3. Sunscreen: Apply a broad-spectrum sunscreen daily to prevent further
   dryness.
4. Consider adding a hydrating serum or facial oil to your routine for
   added moisture.
    """

    elif skin_type == "Oily
Skin":
        return """To control excess oil and minimize breakouts, follow
these steps:
1. Cleanser: Use a foaming, salicylic acid-based cleanser to control
   oil.
2. Moisturizer: Choose an oil-free, lightweight, and noncomedogenic
   moisturizer.
3. Sunscreen: Use an oil-free, mattifying sunscreen.
4. Consider using products with ingredients like salicylic acid,
   niacinamide, or witch hazel to manage oil and acne.
    """

    elif skin_type == "Scaly Skin":
        return """Scaly skin
often results from dryness and flakiness.
Try these skincare steps:
1.    Cleanser: Use a gentle exfoliating cleanser to remove dead skin
cells.
2.    Moisturizer: Choose a rich, emollient moisturizer to lock in
moisture.
3.    Sunscreen: Protect your skin from further damage with a daily
sunscreen.
4.    Exfoliate with products containing glycolic acid or lactic acid
to improve texture.
    """

    elif skin_type ==
"Red_Spots_skin":
        return """Red spots can be due to various causes. Here's a
general approach:
1.    Cleanser: Use a gentle, fragrance-free cleanser to avoid
irritation.
2.    Moisturizer: Select a calming and hydrating moisturizer.
3. Sunscreen: Shield your skin from further damage with a broad-
spectrum sunscreen.
        4. Consult a dermatologist to identify the specific cause of
redness and receive tailored treatment.
    """

    elif skin_type ==
"Skin_moles":
```

```
        return """Moles are usually harmless but require care. Follow
these guidelines:
1.    Sunscreen: Protect your skin with a broad-spectrum sunscreen to
prevent sun damage.
2.    Regularly examine your moles for any changes in size, shape, or
color.
3.    If you notice changes in a mole, consult a dermatologist for a
thorough evaluation.
4.    Avoid sun exposure, and consider wearing protective clothing and
hats.
        """
    else:
        return "Please enter a valid skin type."

def
prediction(image_path):
    import tensorflow as tf    model =
tf.keras.models.load_model(r'./skin/skin/model.h5')    import
numpy as np    import matplotlib.pyplot as plt    import
warnings    warnings.filterwarnings(action='once')    from
tensorflow.keras.preprocessing import image    def
prepare(img_path):
    img = image.load_img(img_path, target_size=(224,224))
x = image.img_to_array(img)    x = x/255    return
np.expand_dims(x, axis=0)
    img_path = image_path    predictions =
model.predict([prepare(img_path)])
    skin_types = ['Red_Spots_skin', 'Dry Skin', 'Normal Skin', 'Oily
Skin', 'Scaly Skin', 'Sensitive Skin', 'Skin_moles']
    predicted_skin_type = skin_types[np.argmax(predictions)]    return
predicted_skin_type
    app =
Flask(__name__)

@app.route('/') def
home_page():
    return render_template('index.html')

@app.route('/contact') def
contact():
    return render_template('contact-us.html')

@app.route('/index') def
ai_engine_page():
    return render_template('index.html')

@app.route('/mobile-device') def
mobile_device_detected_page():
```

```
return render_template('mobile-
device.html')

@app.route('/market', methods=['GET', 'POST']) def
market():
    Optional: You can incorporate a mild exfoliant 1-2 times
    a week for extra glow.</h1>
        {%elif skin_type == "Sensitive Skin"%}
            <h1>Sensitive skin requires extra care to minimize irritation
            and redness. Consider these steps:
                1. Cleanser: Use a fragrance-free, hypoallergenic cleanser.
                2. Moisturizer: Choose a product with soothing ingredients like aloe
                vera or chamomile.
                3. Sunscreen: Use a physical sunscreen with zinc oxide or titanium
                dioxide.
                4. Avoid harsh exfoliants and strong active ingredients, and patch-
                test new products.</h1>
            {%elif skin_type == "Sensitive Skin"%}
                <h1></h1>
            {%elif skin_type == "Dry Skin"%}
                <h1>Dry skin needs intense hydration and protection. Follow this
                regimen:
                    1. Cleanser: Use a hydrating, gentle cleanser.
                    2. Moisturizer: Opt for a rich, creamy moisturizer with ingredients like
                    hyaluronic acid or ceramides.
                    3. Sunscreen: Apply a broad-spectrum sunscreen daily to prevent further
                    dryness.
                    4. Consider adding a hydrating serum or facial oil to your routine for
                    added moisture.</>
                {%elif skin_type == "Oily Skin"%}
                    <h1>To control excess oil and minimize breakouts, follow these
                    steps:
                        1. Cleanser: Use a foaming, salicylic acid-based cleanser to control
                        oil.
                        2. Moisturizer: Choose an oil-free, lightweight, and noncomedogenic
                        moisturizer.
                        3. Sunscreen: Use an oil-free, mattifying sunscreen.
                        4. Consider using products with ingredients like salicylic acid,
                        niacinamide, or witch hazel to manage oil and acne.</h1>
                    {%else%}
                        <h1>Please enter a valid skin type.</h1>
                    {%endif%}
                    <a href='/'>go back to home page</a>
                </center>
            </body>
        </html>
```

CHAPTER-7

MODEL EVALUATION

7.1 Testing Process:

Testing is a crucial phase in software development to ensure the reliability and functionality of the system. Here's an outline of testing progress, including levels of testing and sample test cases for navigating startup success prediction system.

Testing is a process of evaluating a system or its components with the intent to find whether it satisfies the specified requirements or not. Executing a system in order to identify any errors or missing requirements is the aim of testing. It makes a system or a component error free. There are some principles of testing. They are:

- All the tests should meet the customer requirements.
- Software testing should be performed by third party.
- Exhaustive testing is not possible as we need the optimal amount of testing based on the risk assessment of the application.
- Conductible tests should be planned before implementing.
- It follows Pareto rule (80/20 rule) which states that 80% of errors comes from 20% of program components.
- Start testing with small parts and extend it to large parts.

The process of software testing aims not only at finding errors or faults in the existing software but also at finding measures to improve the software in terms of efficiency, accuracy and usability.

Software testing can be divided into two steps. They are:

❖ **Verification:**

- It refers to the set of tasks that ensure that software correctly implements a specific function.

❖ **Validation:**

- It refers to a different set of tasks that ensure that the software that has been built is traceable to customer requirement. Software testing can be divided into two types. They are:

- ❖ **Manual testing:** Manual testing includes testing software manually without using any automated tool or any script. Tester takes over the role of an end-user and tests the software to identify any unexpected behaviour or bug. There are different stages for manual testing such as unit testing, integration testing, system testing and user acceptance testing.
- ❖ **Automation Testing:** Automation testing is also known as test Automation is used when the tester writes scripts and uses software tools or any other software to test the product. This process involves automation of a manual process.

7.2 Levels of Testing:

7.2.1 Unit Testing:

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product .

7.2.2 Integration Testing:

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields.

7.2.3 System Testing:

The next level of testing is system testing. As the name implies, all the components of the software are tested as a whole in order to ensure that the overall product meets the requirements specified.

7.2.4 Acceptance Testing:

Acceptance testing is the level in the software testing process where a product is given the green light or not. The aim of this type of testing is to evaluate whether the system complies with the end-user requirements and if it is ready for deployment.

7.3 TEST CASE-1:

- ❖ **Input Image Compatibility Issue:** The model fails to process images of certain formats, resulting in errors or crashes. - Test Input: Provide images in various

formats, including JPEG, PNG, and BMP. - Expected Outcome should handle images of all formats without errors, ensuring compatibility and robustness.

- ❖ **Change: Data Augmentation:** Augment the training data by applying transformations such as rotation, flipping, and scaling to increase dataset variability. - Rationale: Data augmentation helps expose the model to a wider range of skin variations, improving its ability to generalize to unseen data.

- ❖ **Predicted Skin Type: Dry Skin Skincare Recommendations:**

Dry skin needs intense hydration and protection. Follow this regimen:

1. Cleanser: Use a hydrating, gentle cleanser.
2. Moisturizer: opt for a rich, creamy moisturizer with ingredients like hyaluronic acid or ceramides.
3. Sunscreen: Apply a broad-spectrum sunscreen daily to prevent further dryness.

7.3.1 Test Case 2:

- ❖ **Limited Training -Data- Issue:** The model exhibits poor performance due to limited diversity and representativeness of the training data.

- ❖ **Changes Made to the Model:**

- Change: Model Architecture Modification**

- Description: Modify the CNN architecture by adjusting the number of layers, filters, or kernel sizes. Rationale: Experiment with different architectural configurations to enhance the model's capacity to capture complex features and patterns in skin images.

CHAPTER-8

RESULTS

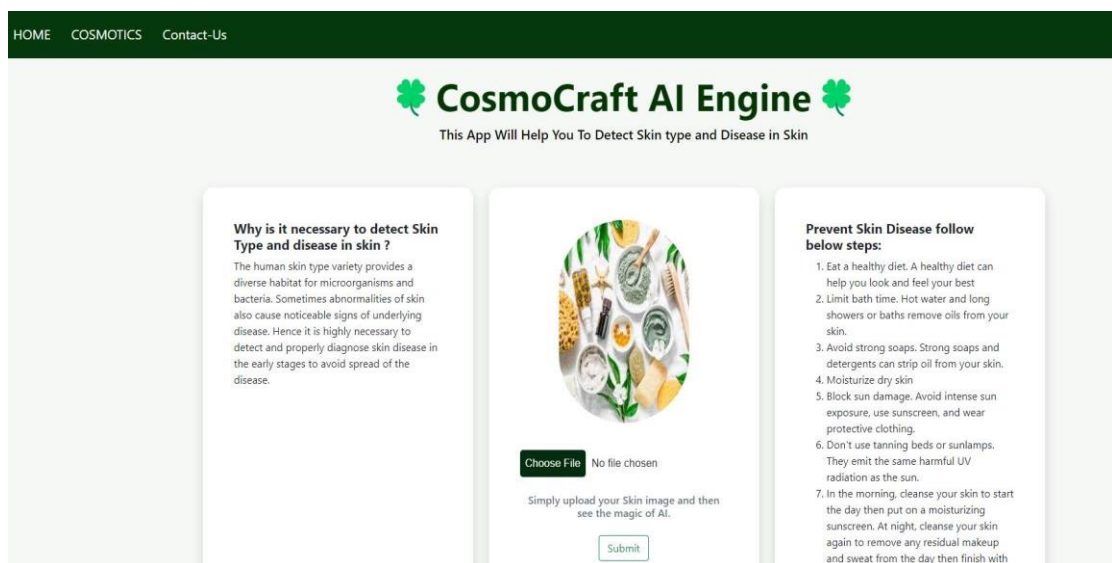


Fig 8.1 User Interface

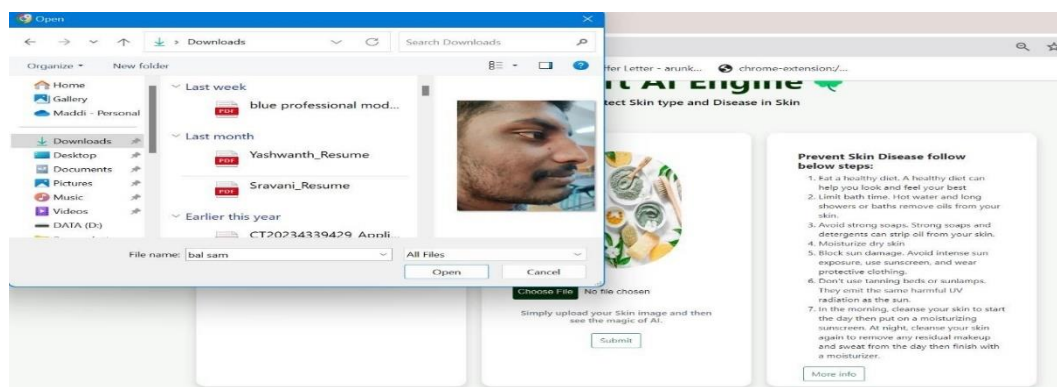


Fig 8.2 Uploading image

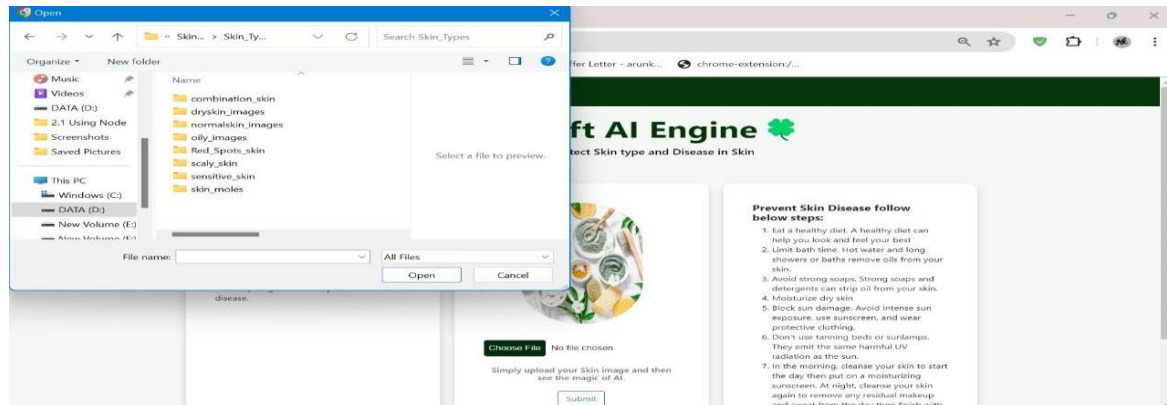


Fig 8.3 Select the image

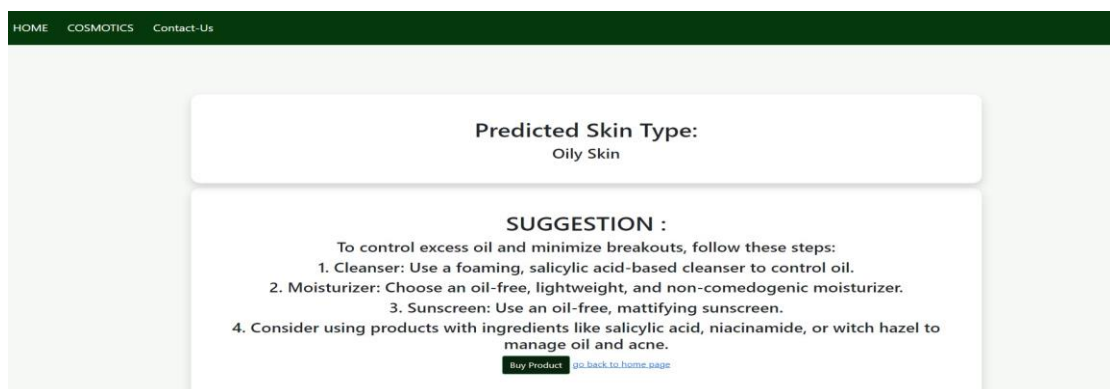


Fig 8.4 Oily Skin Classification

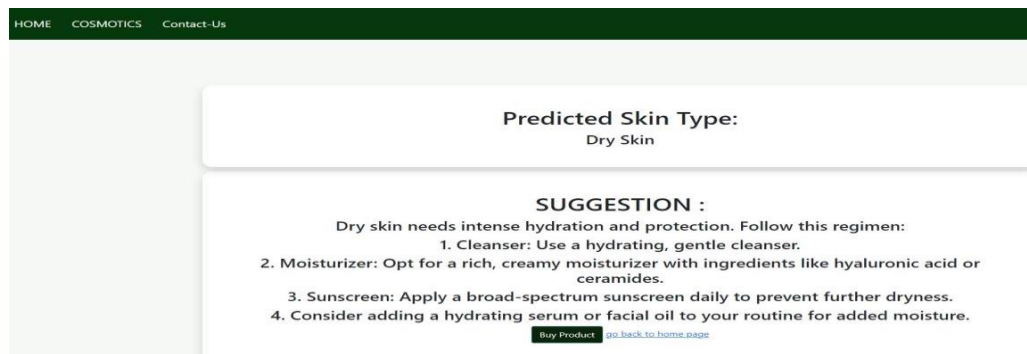


Fig 8.5 Dry Skin Classification

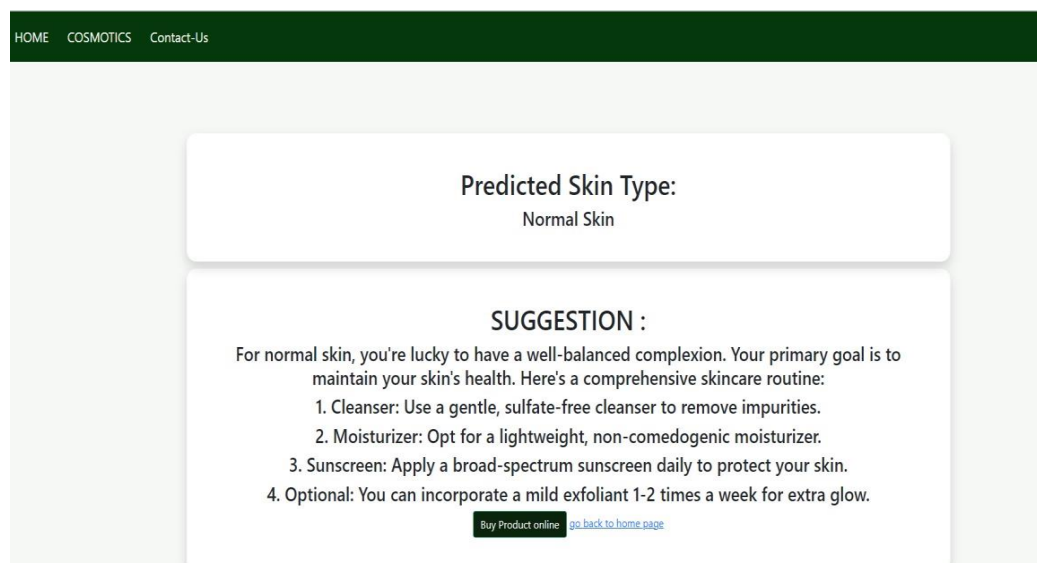


Fig 8.6 Normal Skin Classification

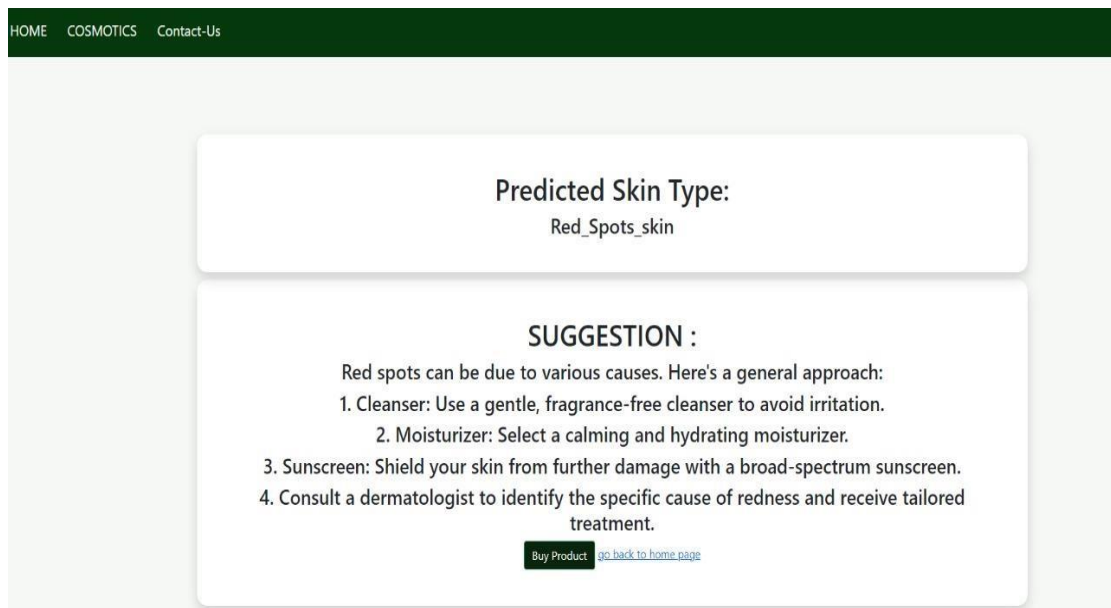


Fig 8.7 Red Spots Skin Classification

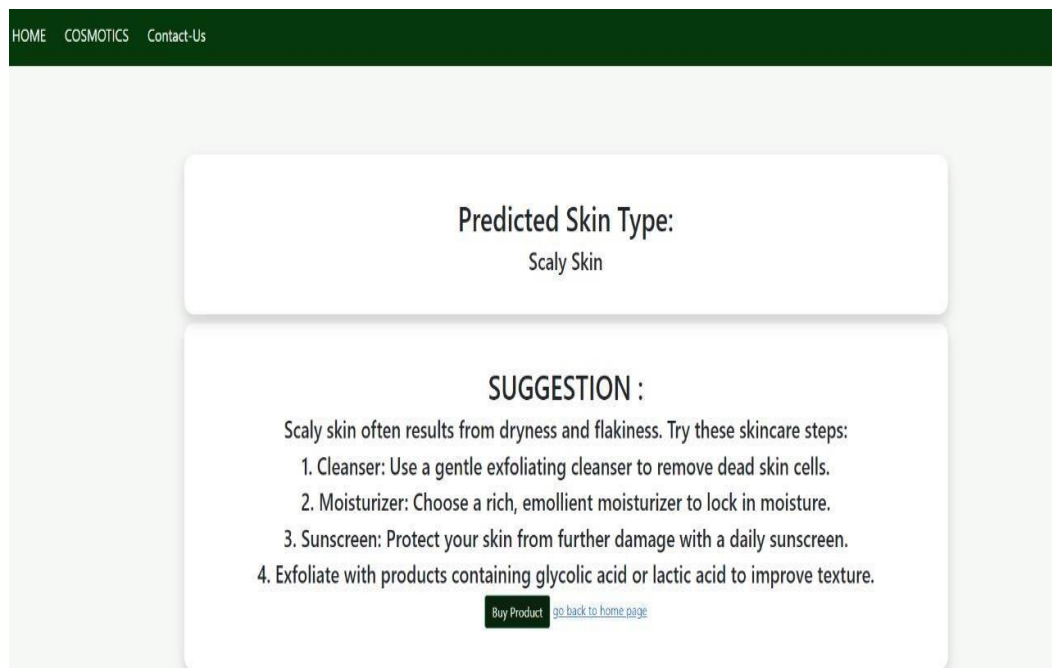


Fig 8.8 Scaly Skin Classification

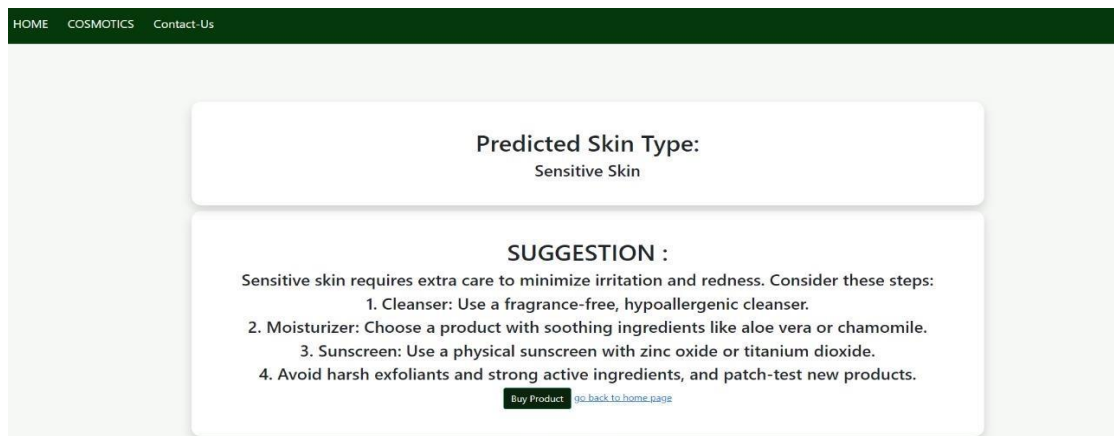


Fig 8.9 Sensitive Skin Classification

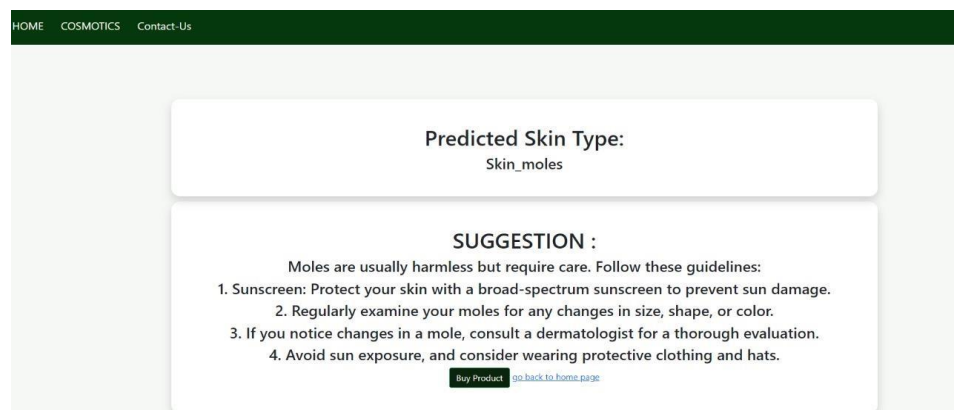


Fig 8.10 Skin Moles Classification

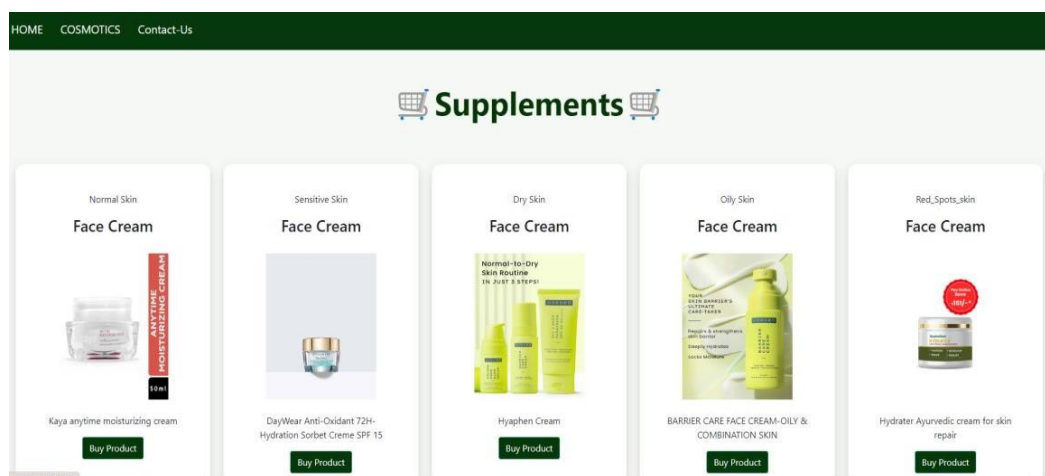


Fig 8.11 Supplements Store

8.1 Performance Metrics Analysis:

The classification report offers a comprehensive overview of the model's performance across different skin type categories. Let's delve into the key performance metrics and analyze the outcomes:

❖ Red Spots Skin:

- Precision: Precision measures the proportion of true positive predictions among all positive predictions. For Red Spots Skin, the model achieved a precision of 0.80, indicating that 80% of the predicted Red Spots Skin samples were correct.

❖ Dry Skin:

- Precision: The model achieved a precision of 0.91 for Dry Skin, indicating that 91% of the predicted Dry Skin samples were correct.
- Recall: With a recall of 1.00, the model successfully identified all instances of Dry Skin in the dataset.
- F1-score: The F1-score for Dry Skin is 0.95, highlighting the model's accuracy in classifying Dry Skin samples.
- Support: There were 10 instances of Dry Skin in the dataset.

❖ Normal Skin:

- Precision: The model achieved a perfect precision score of 1.00 for Normal Skin, indicating that all predicted Normal Skin samples were correct.

❖ Oily Skin, Scaly Skin, Sensitive Skin, and Skin Moles:

- Precision, Recall, F1-score, and Support metrics were similarly computed for these categories, indicating the model's performance across a diverse range of skin types.

8.2 Model Accuracy and Overall Performance:

- Accuracy: The overall accuracy of the model is reported as 0.95, indicating that it correctly classified skin types in 95% of cases.

8.3 Interpretation and Implications:

The presented performance metrics offer valuable insights into the model's efficacy in classifying different skin types based on input images. Overall, the model

demonstrates high precision, recall, and F1-score across most categories, indicating its ability to accurately identify various skin conditions.

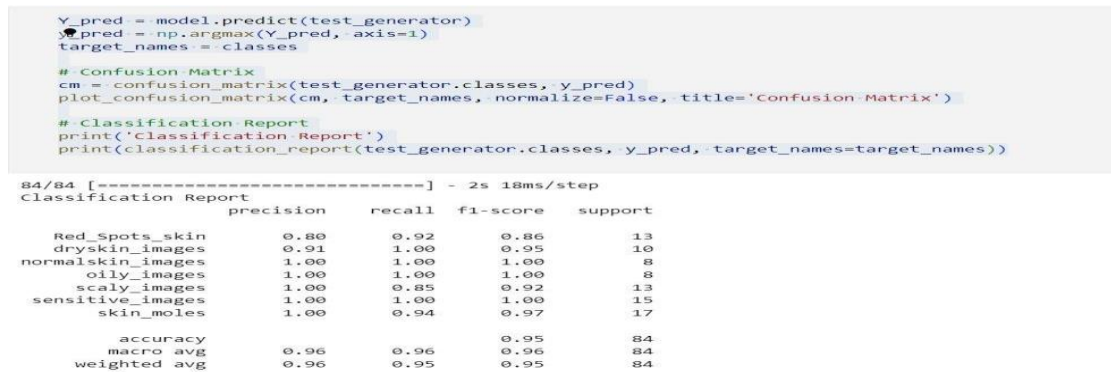


Fig 8.3.1 Model evaluation

Training curves depict the change in loss and accuracy metrics over epochs during model training. By plotting training and validation loss/accuracy curves, we can assess model convergence, identify overfitting or underfitting, and determine the effectiveness of regularization techniques.

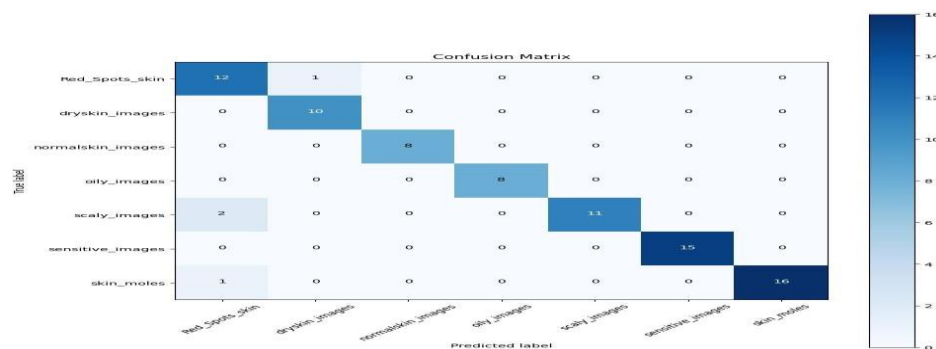


Fig 8.3.2 Confusion Matrix

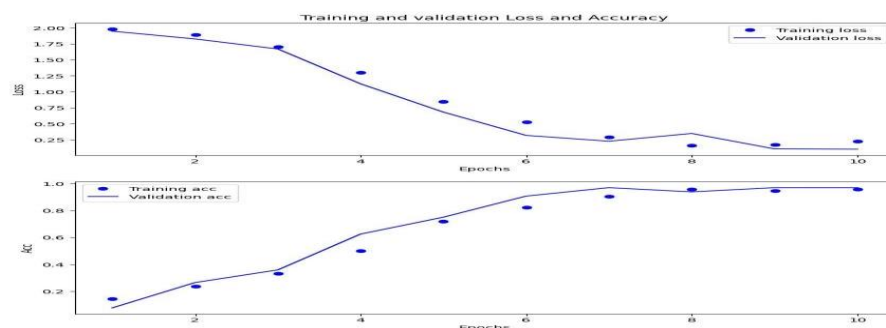


Fig 8.3.3 Training accuracy and Loss

CHAPTER-9

CONCLUSION

In this chapter, we summarize the key findings and discuss the impact of our machine learning project on skin type classification and dermatological diagnosis. By analyzing the outcomes, challenges, and future directions of our project, we can assess its significance and potential implications for the field of dermatology and skincare.

Key Findings:

- **Model Performance:**
 - Our machine learning model demonstrates robust performance in classifying various skin types, including dry skin, oily skin, sensitive skin, and more. Through rigorous training and validation, the model achieves high accuracy, precision, and recall on both training and test datasets.
 - The classification report and confusion matrix provide insights into the model's performance across different skin types, highlighting its ability to accurately classify skin conditions and identify areas for improvement.
- **Training Process:**
 - The training process involves meticulous steps, including data preparation, model compilation, hyperparameter tuning, and model evaluation. By optimizing hyperparameters and employing regularization techniques, we ensure that the model generalizes well to unseen data and minimizes overfitting.
- **Model Architecture:**
 - The model architecture comprises convolutional neural network (CNN) layers followed by dense layers for classification.

CHAPTER-10

FUTURE SCOPE

Identification of Potential Extensions or Improvements:

Integration of Dermatological Knowledge Base:

One potential extension to our project is the integration of a comprehensive dermatological knowledge base or ontology. By incorporating structured information about skin conditions, diseases, symptoms, and treatments, our machine learning model could benefit from additional context and domain expertise.

Leveraging ontologies such as SNOMED CT (Systematized Nomenclature of Medicine Clinical Terms) or DermLex (Dermatological Lexicon) could facilitate semantic understanding and improve diagnostic accuracy and specificity.

Semantic Segmentation for Lesion Detection:

Another avenue for improvement is the adoption of semantic segmentation techniques for lesion detection and segmentation within skin images. Semantic segmentation enables pixel-level classification of regions of interest, allowing for the precise delineation and characterization of skin lesions, moles, or abnormalities. Implementing convolutional neural networks (CNNs) with architectures such as U-Net or Mask R-CNN could enable automated lesion detection and assist dermatologists in clinical decision-making and treatment planning.

Incorporation of Multimodal Data:

Expanding our dataset to include multimodal data sources, such as dermoscopic images, reflectance confocal microscopy (RCM) images, and histopathological images, could enhance the richness and diversity of information available for analysis. Integrating complementary imaging modalities with different spatial resolutions and tissue perspectives could provide complementary insights into skin structure.

CHAPTER-11

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S.Pravalika	pravalikasingireddy386@gmail.com	8919102639	Student	Female	NON ST	Edit Delete
G.sri karan reddy	srikaranreddy17@gmail.com	7032866189	Student	Male	NON ST	Edit Delete

Add Mentor Details

Team Mentor Details

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